

Modern Knowledge Representation: very (VERY) large scale, and how to combine KR with ML

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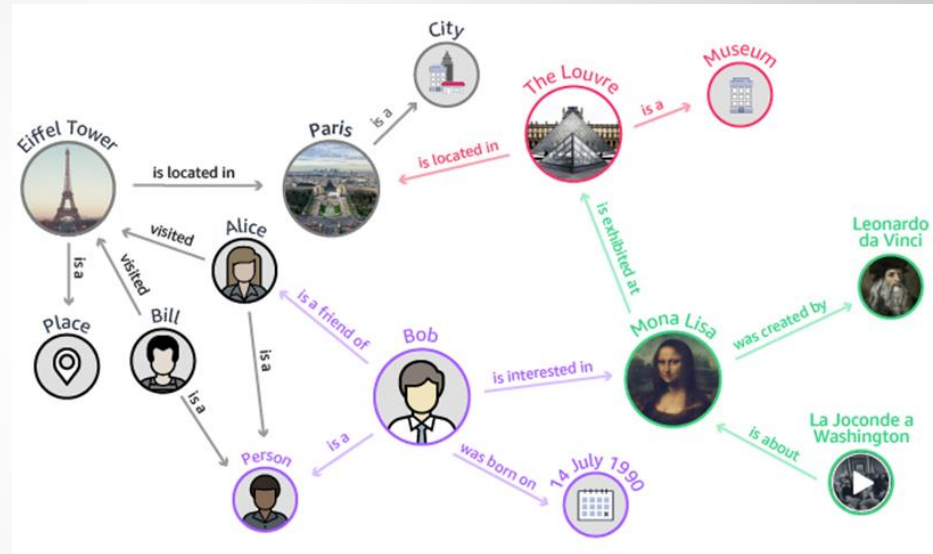
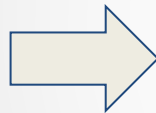


Handbook of Knowledge Representation (1000 pages, ToC alone is 14 pages)

- propositional logic & satisfiability solvers
- first order logic & resolution
- description logic
- constraint (logic) programming
- nonmonotonic reasoning
- belief revision
- qualitative reasoning
- model-based diagnosis
- bayesian networks
- temporal logic
- spatial reasoning
- epistemic logic
- deontic logic
- situation calculus
- default logic
- event calculus
- **knowledge graphs**
-

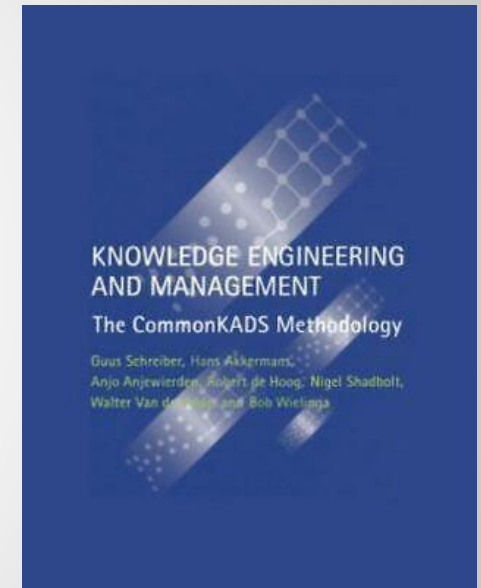
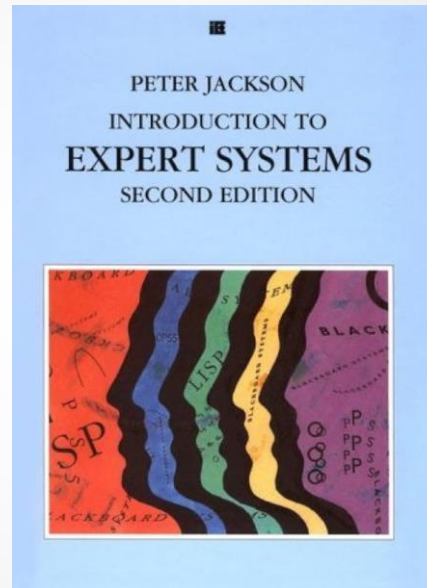
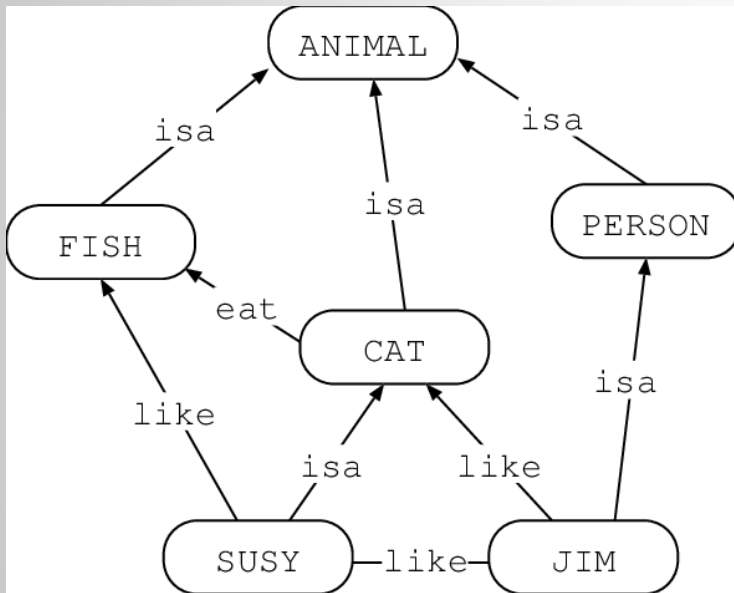


Knowledge Graphs and more



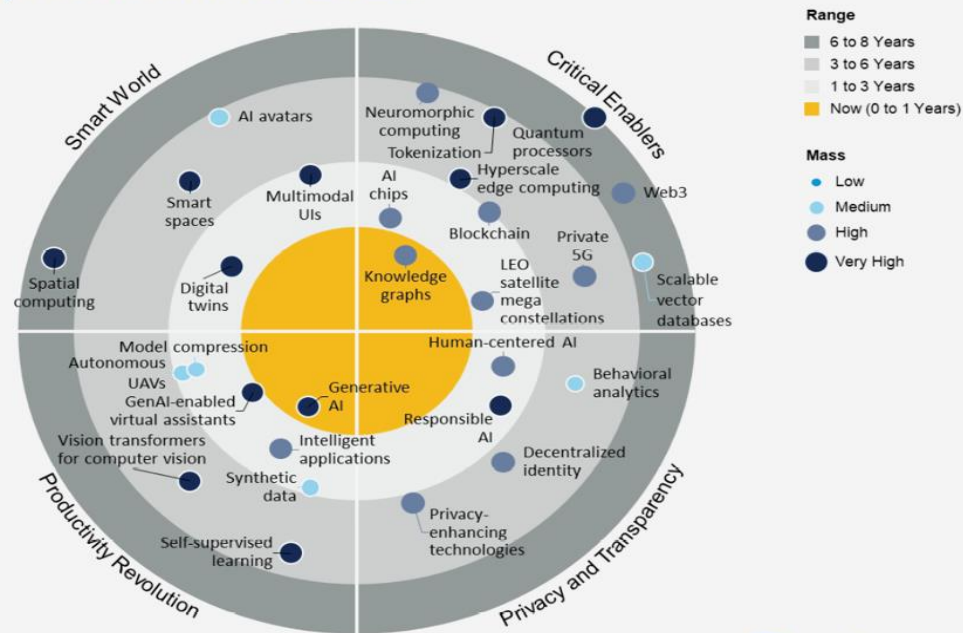
Knowledge graphs (KGs) are machine-readable data structures that describe the relationship between heterogeneous data via a network of nodes and links.

Back to the 90s?



Back to the Future!

Impact Radar for 2024

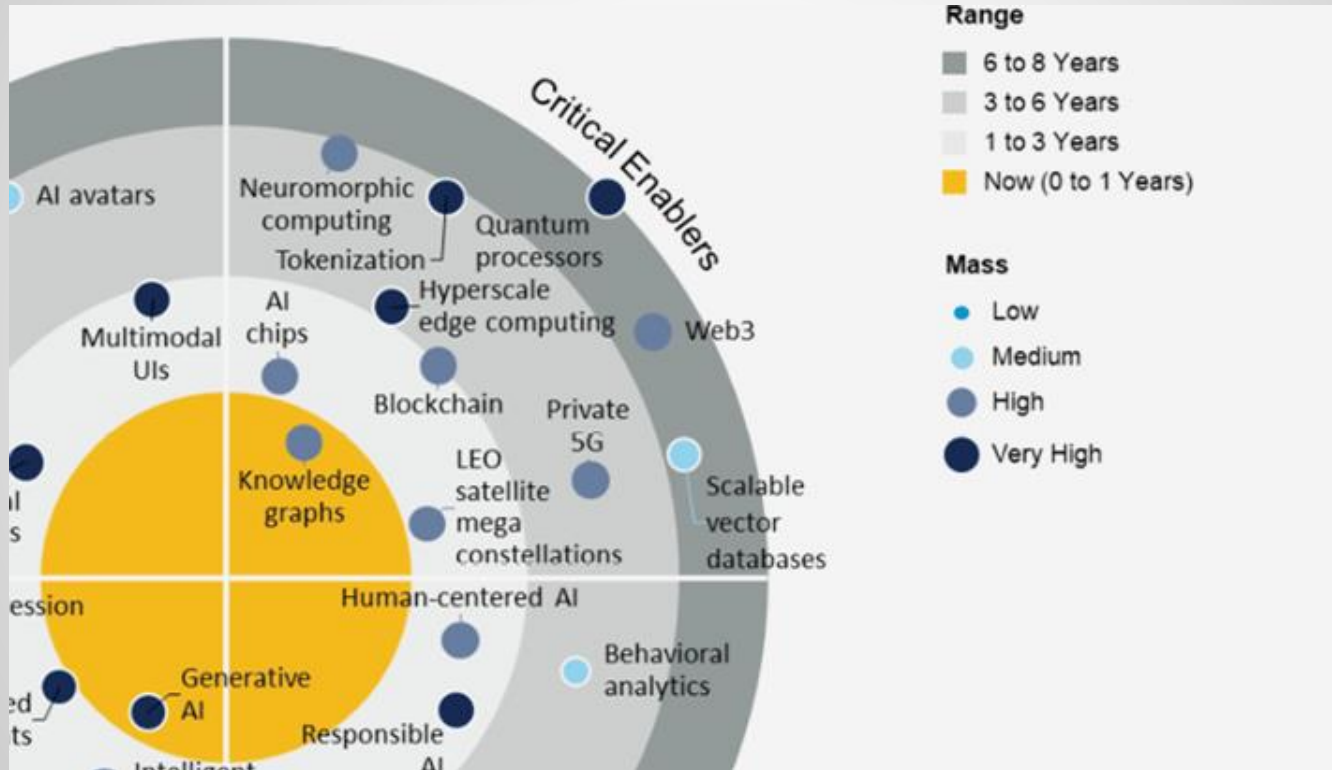


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Gartner

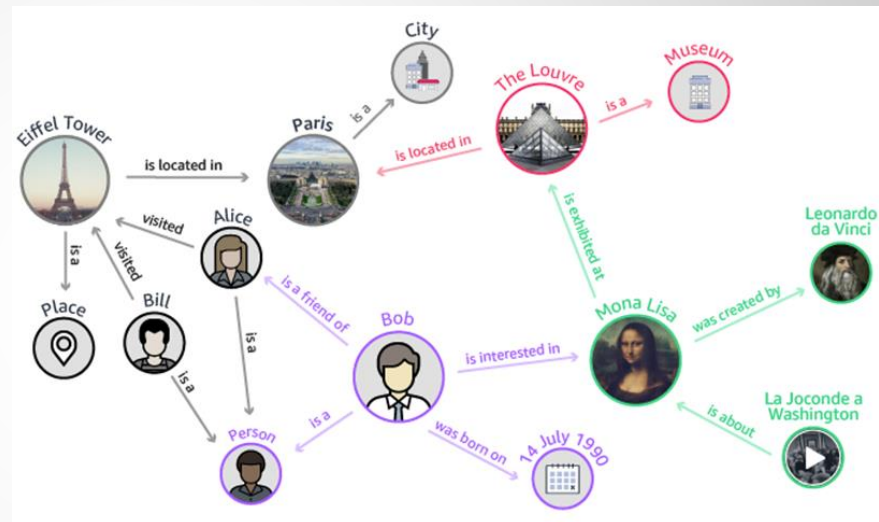
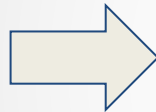
<https://www.gartner.com/en/articles/30-emerging-technologies-that-will-guide-your-business-decisions>

Back to the Future!



<https://www.gartner.com/en/articles/30-emerging-technologies-that-will-guide-your-business-decisions>

Knowledge Graphs, Ontologies and Logics



Knowledge graphs (KGs) are machine-readable data structures that describe the relationship between heterogeneous data via a network of nodes and links.

Additional assumption: nodes and links are **labelled** and come with **Semantics** (formal, social)

Knowledge Graphs, Ontologies and Logic

Knowledge graphs (KGs) are machine-readable *data structures* that describe the relationship between heterogeneous data via a network of nodes and links.

Ontologies are shared *conceptualisations* of the world, often w.r.t. specific domain and application, and ideally formalised and machine readable.

Logics are *formal systems* (often machine readable) that allow us to represent knowledge, and to (automatically) reason over it according to well-understood semantics and computational properties

Knowledge Representation and Knowledge Engineering

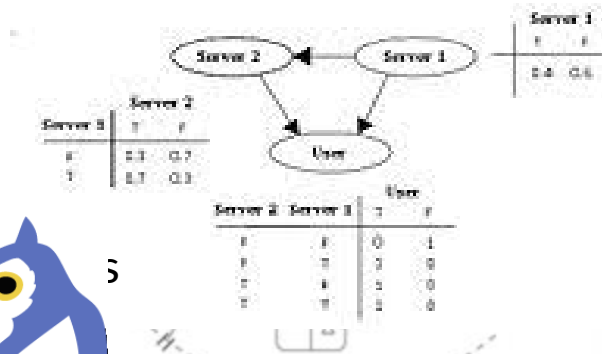
Knowledge Representation is about formal/explicit *representations*:

- to represent knowledge as **surrogates** (modeling the world)
- to agree on joint world views (**ontological commitment**)
- to facilitate (automated and systematic) **intelligent reasoning**
- with **efficient computation** (or at least well understood)
- act as proxies for **human expressions**

Knowledge Engineering is about:

- **constructing** formal representations, and
 - covers the entire **lifecycle** of knowledge in applications
 - related to Software Engineering
-

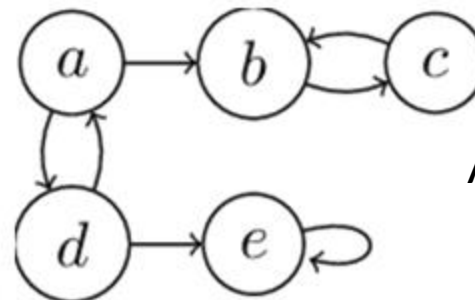
But not all Knowledge is Knowledge Graphs



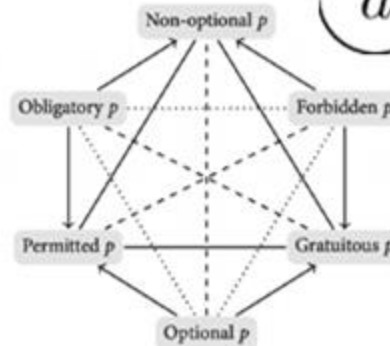
S

Syntax	Description
A, B	Atomic concept
C, D	Concept descriptions
R	Atomic Role
T	Universal concept
⊥	Bottom concept
¬A	Atomic negation
C ⊓ D	Concept intersection
∀ R. I	Value restriction
∃ R. T	Limited existential quantification

S



Arguments

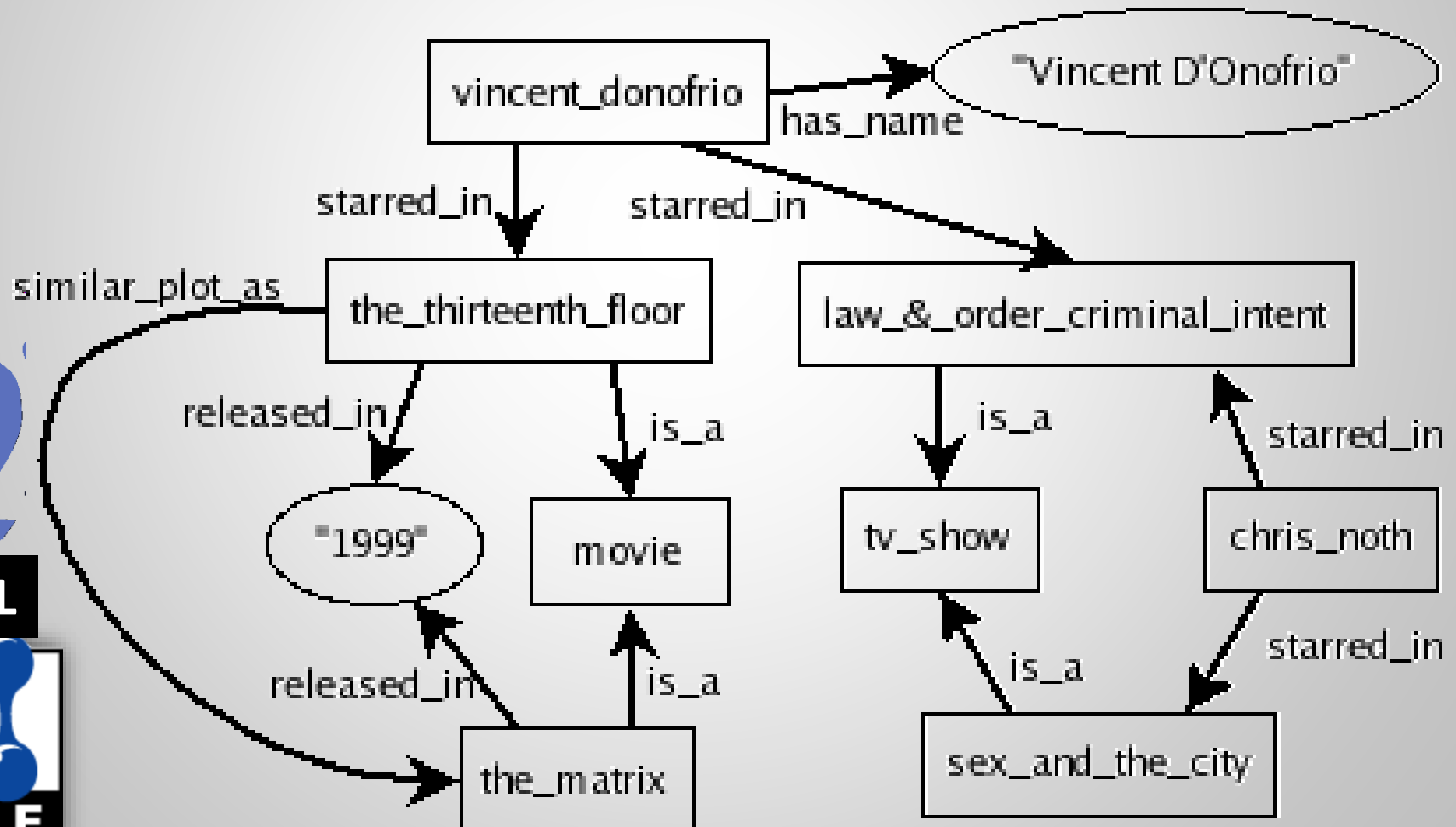


and many more...

Very (VERY) large scale KR:
Knowledge Graphs

What are Knowledge Graphs

Most widely used KR formalism in history
(by many orders of magnitude)



OWL



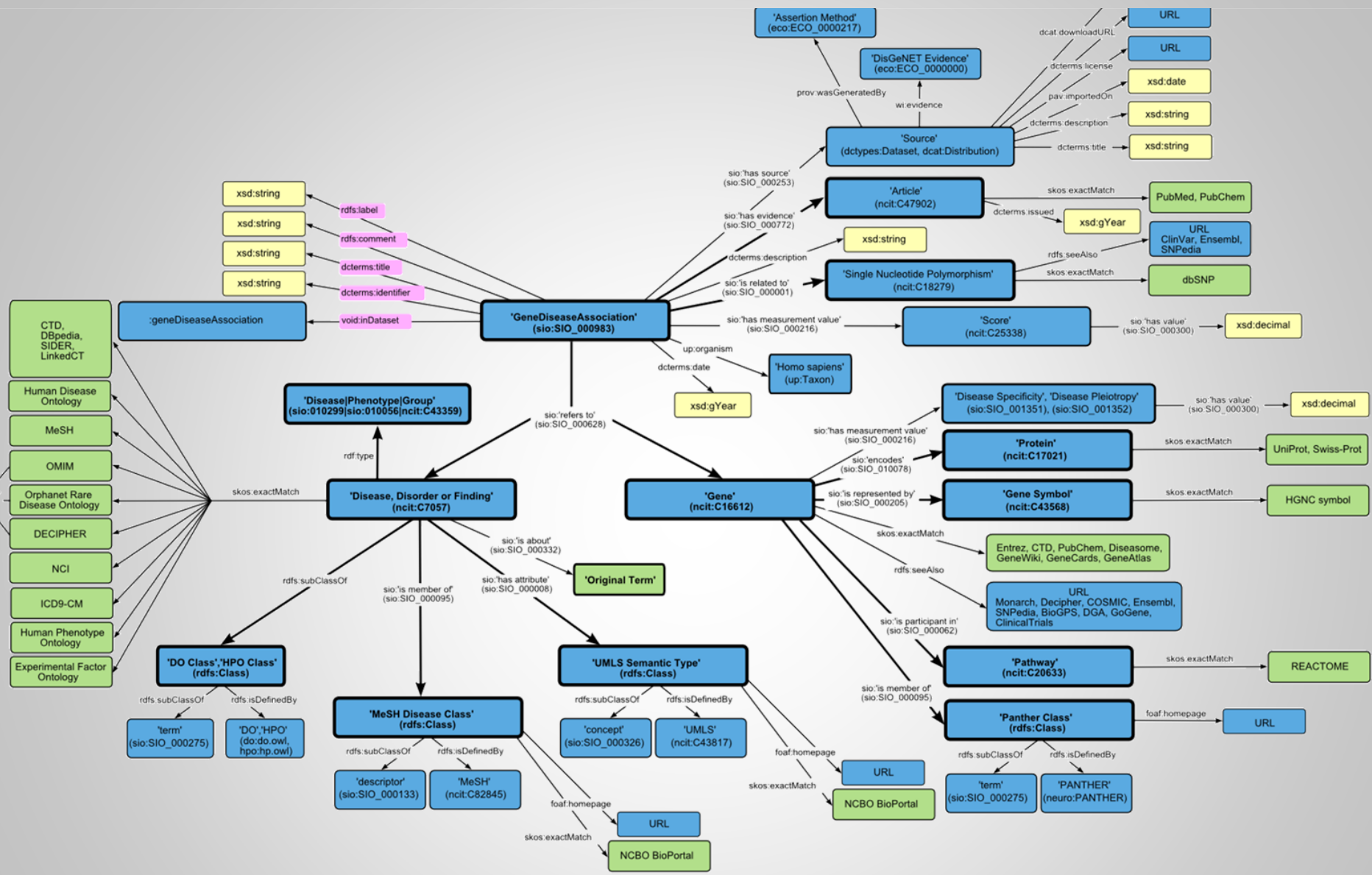
RDF

What are Knowledge Graphs (for logicians)

- ground binary predicate: $P(O_1, O_2)$
- Limited existential variables: $\exists x: P(C_1, x) \wedge P(C_2, x)$
- Types is unary predicate: $T_i(x)$
- Subtypes $\forall x: T_1(x) \rightarrow T_2(x)$
- Type restrictions $\forall x, y: P(x, y) \rightarrow T_1(x) \wedge T_2(y)$
- Equality: $O_1 = O_2$
- Extensions to **DL**:
 - Disjointness of types
 - Cardinality restrictions (0,1)
 - always decidable: sub-FOL.

Logical deduction

What's in these knowledge graphs,
and how big do they get?

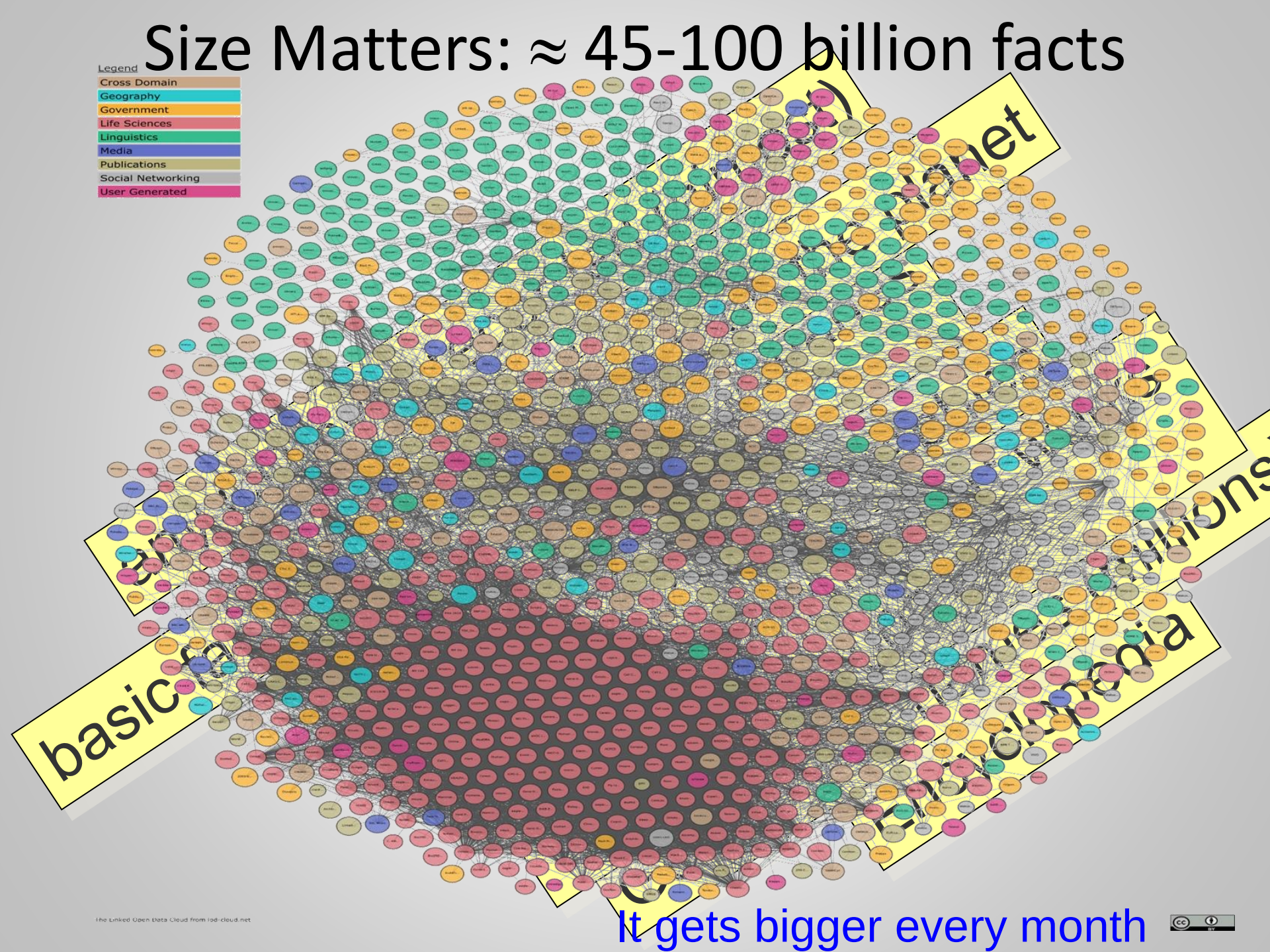


disgenet.org: 45.000 SNP, 17.000 genes, 15.000 diseases

Size Matters: $\approx 45\text{-}100$ billion facts

Legend

Cross Domain
Geography
Government
Life Sciences
Linguistics
Media
Publications
Social Networking
User Generated



How big are Knowledge Graphs?

A white golf ball with a dimpled texture is positioned on a white tee, set against a background of green grass and a blue sky with a bright sun. The text "1 fact" is overlaid on the center of the golf ball.

1 fact







Suez Canal

10⁷ Facts



Moon

10⁸ Facts



Earth

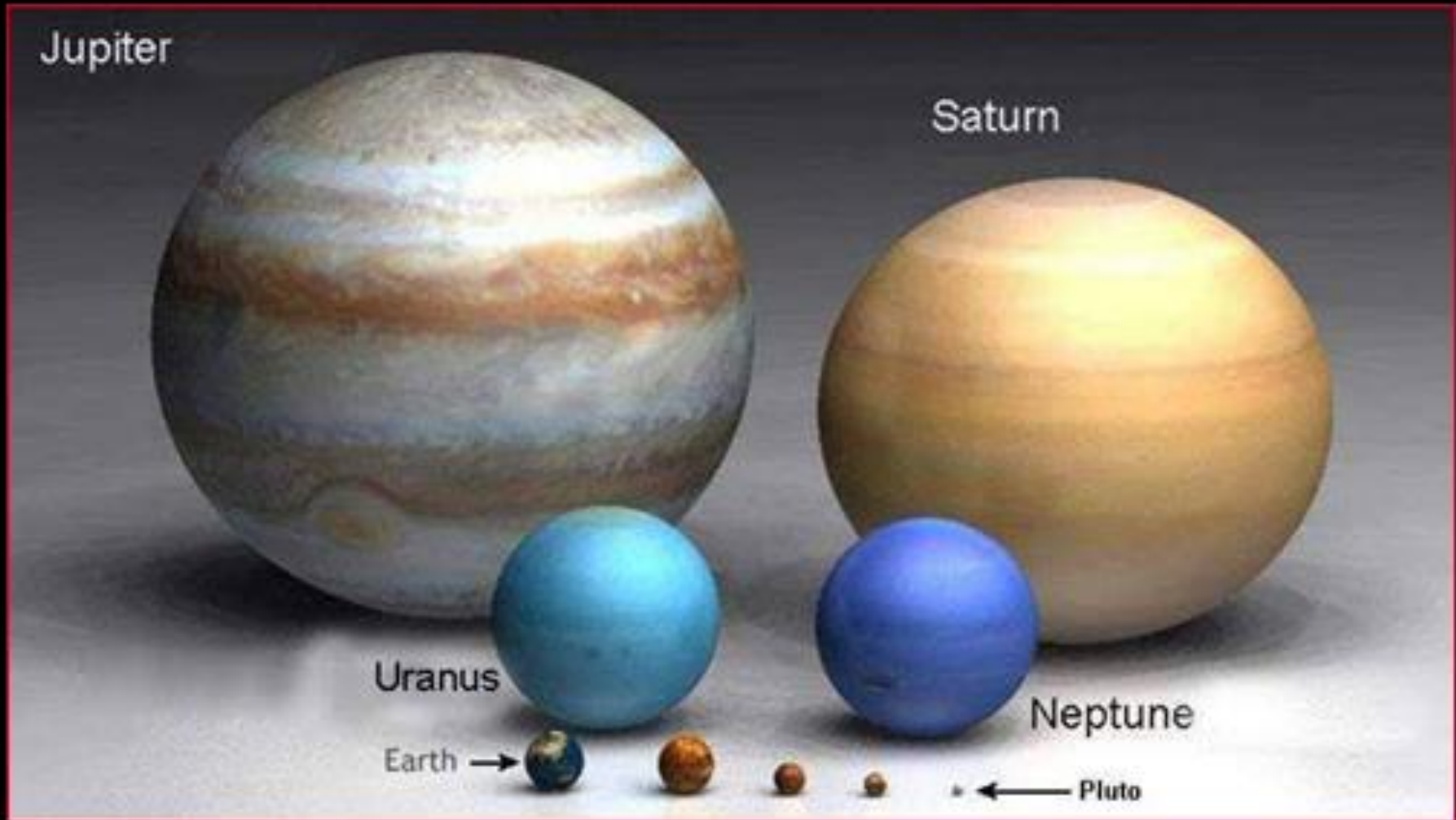
$\sim 10^9$ Facts



Size of the current Semantic Web

$\sim 10^{10}$ Facts

Jupiter



≈ 1 fact per web-page

Is anybody using this stuff?

Tencent 腾讯

UniProt USGS

Google

Bing

Alibaba.com

Baidu 百度

PubMed IBM

Cochrane

EMBL-EBI

facebook

Uber

ANTONI VAN LEEUWENHOEK FOUNDATION

Pinterest

airbnb

LinkedIn

MAASTRO

BBC

The New York Times

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NXP nexperia

NETFLIX



REUTERS

Overheid.nl

LEIT

BIBLIOTHEKET

LIBRARY OF CONGRESS

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Bloomberg

EPA United States Environmental Protection Agency

{BnF

National Library of Sweden



Bodleian Libraries UNIVERSITY OF OXFORD

RENAULT

legislation.gov.uk

kadaster

POLITIE

European Commission

Wolters Kluwer

IOS Press

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accenture

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[Utrecht.nl - Homepage](#)

[www.utrecht.nl/](#) - Translate this page

Utrecht de gemeentesite. Welkom op de website van de Gemeente **Utrecht**. Hier vindt u alle informatie over de stad en de diensten van de Gemeente **Utrecht**.

[Loket Burgerzaken - Contact - Afspraak maken Burgerzaken - Milieu en afval](#)

[Utrecht - Wikipedia, the free encyclopedia](#)

[en.wikipedia.org/wiki/Utrecht](#)

Utrecht city and municipality is the capital and most populous city of the Dutch province of **Utrecht**. It is located in the eastern corner of the Randstad conurbation, ...

[Utrecht University - Treaty of Utrecht - Utrecht - DOS](#)

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Welkom in **Utrecht**! **Utrecht**, het kloppende hart van Nederland. Een stad met een unieke dynamiek. Klein genoeg om wandelend de middeleeuwse binnenstad ...

Knowledge Discovery

Predict new uses for existing drugs,

Predict new chemical compounds for new drugs

hide sidebar

REPURPOSING DRUGS WITH SEMANTICS

melanoma - efo/EFO_0000756 (23 connections)

Add Search

Type: ☒ Drug ☐ Disease ☐ Protein

Found 89 connections.

Explore Legend

Node Details Edge Details

Options Help

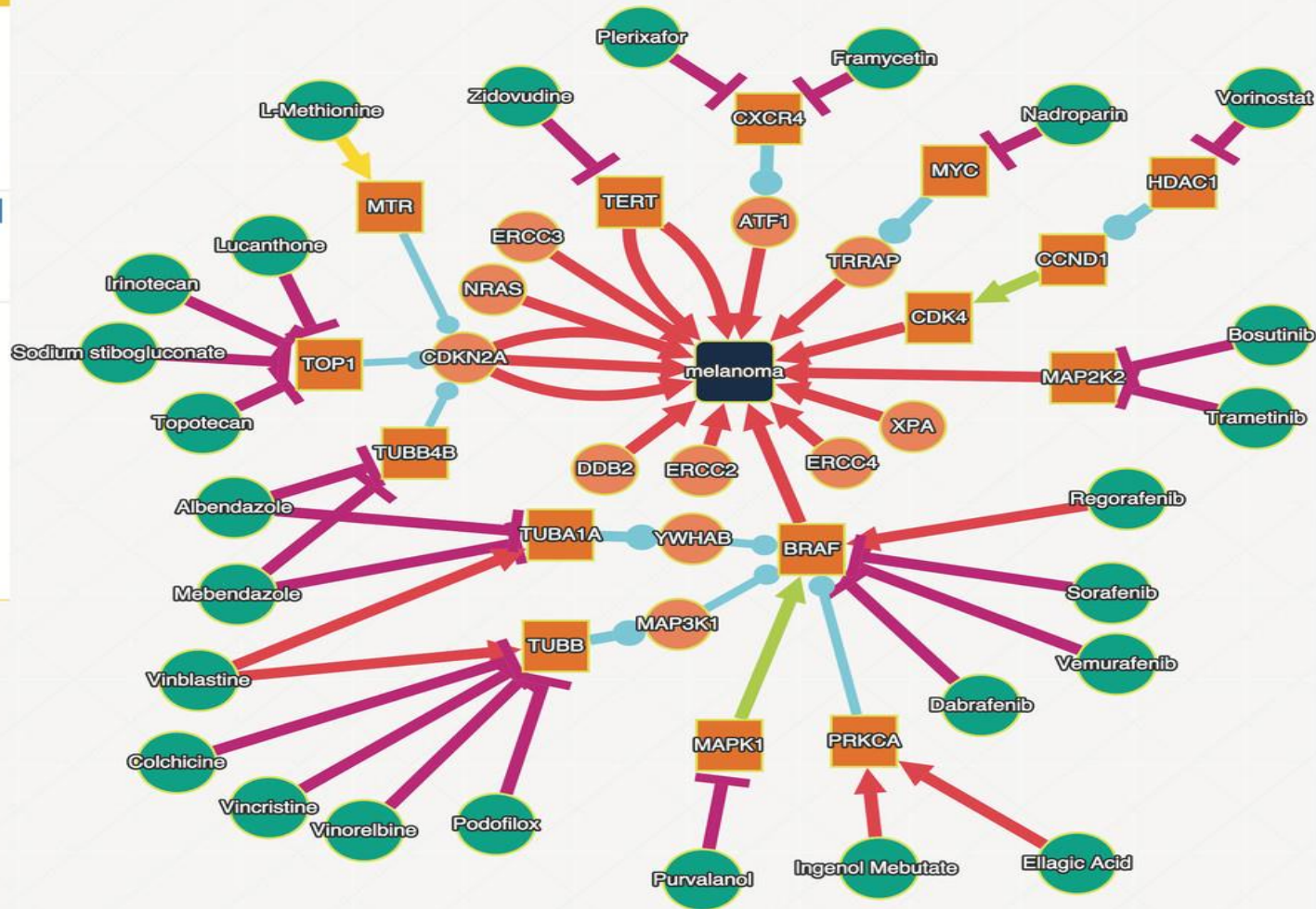
About

NODES:

Protein	Disease
Drug	Undefined

EDGES

Activation	Inhibition
Physical Association	Phosphorylation Reaction
Cleavage	Interaction



SPORT OLYMPICS

The Olympic Broadcaster
London 2012

Home | Football | Formula 1 | Cricket | Rugby Union | Olympics | The Ashes | More sports ▾

Countries | Athlete | Britta Steffen

Schedule & Results | Medals | More Olympic sports ▾

Britta Steffen



Biography

SWIMMING

Britta Steffen (born 16 November 1983 in Schwedt, Germany) is a German swimmer who specializes in freestyle, who is the current holder of the world record in women's 50 and 100 metre freestyle.

[Full biography ▾](#)

Olympic schedule

**414 days to go**

Olympic Games 27 July-12 Aug 2012

[View full olympic schedule](#)

Key facts

Country
GermanyCompetes in
Womans 50m
Freestyle
Womans 100m
FreestyleTeam
Germany Women's
4 x 100m Freestyle
RelayDate of birth
03/03/83Height
1.88m/6'

WOMEN'S 100M FREESTYLE

Britta Steffen sets
new record 100m

Jamaican sprint superstar Usain Bolt claims yet another Olympic record with a stunning victory in the 100m heats.

Olympic venues

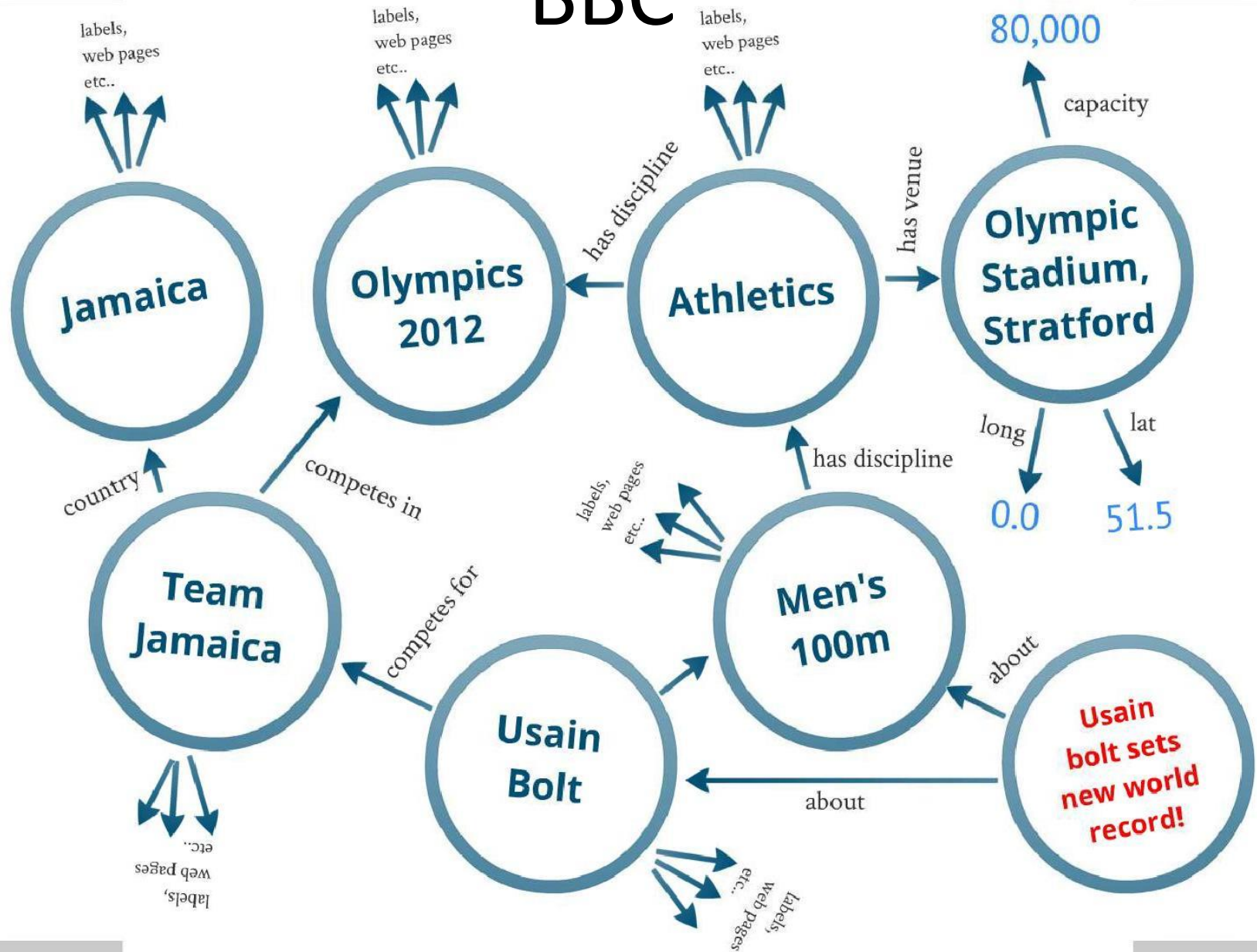


From the 27th of July 2012, 17,000 athletes will compete in 31 venues in the Olympic Park, across London and at out of London venues

[View all London 2012 venues](#)

Olympic 2012 programmes

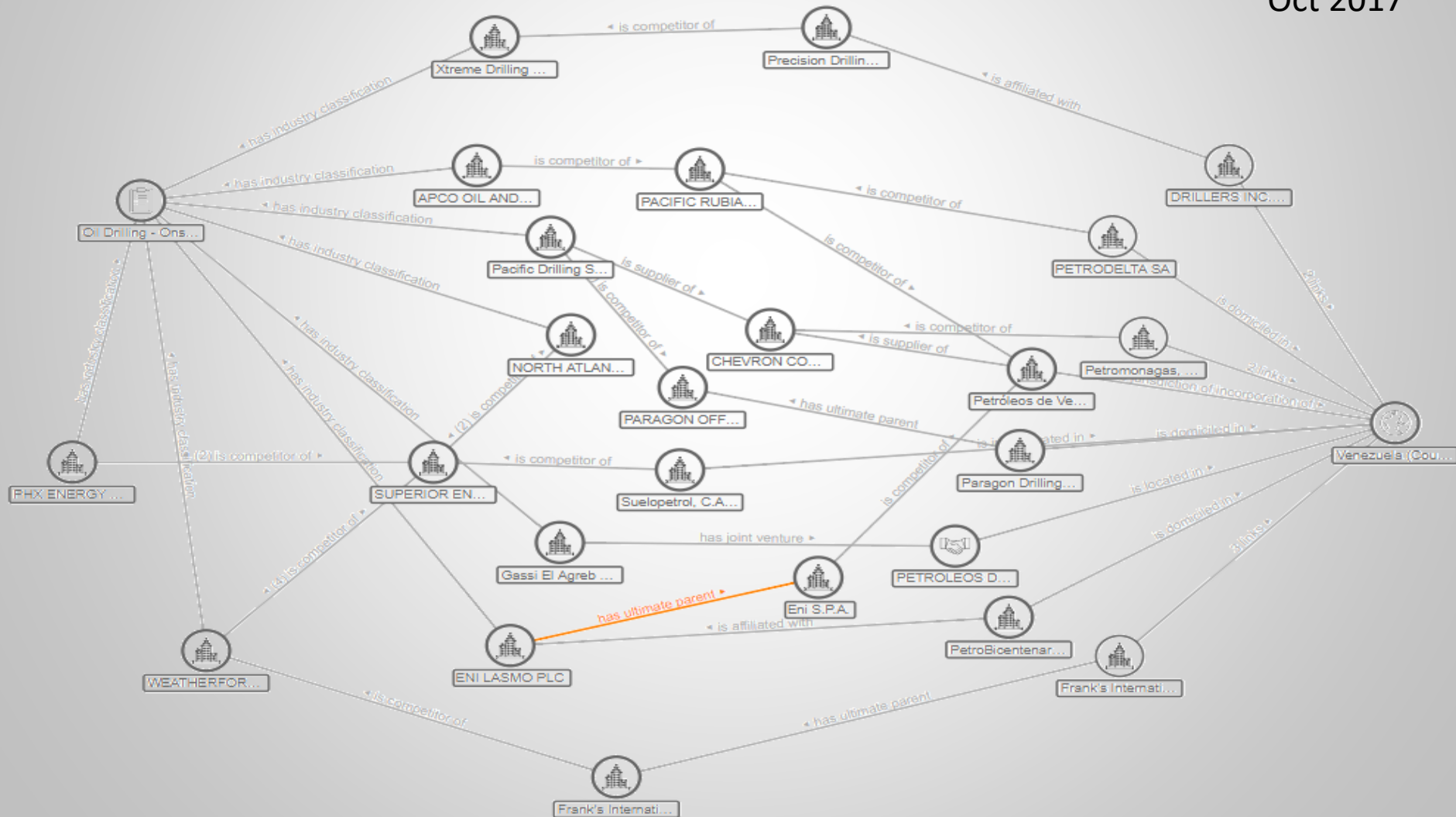
BBC



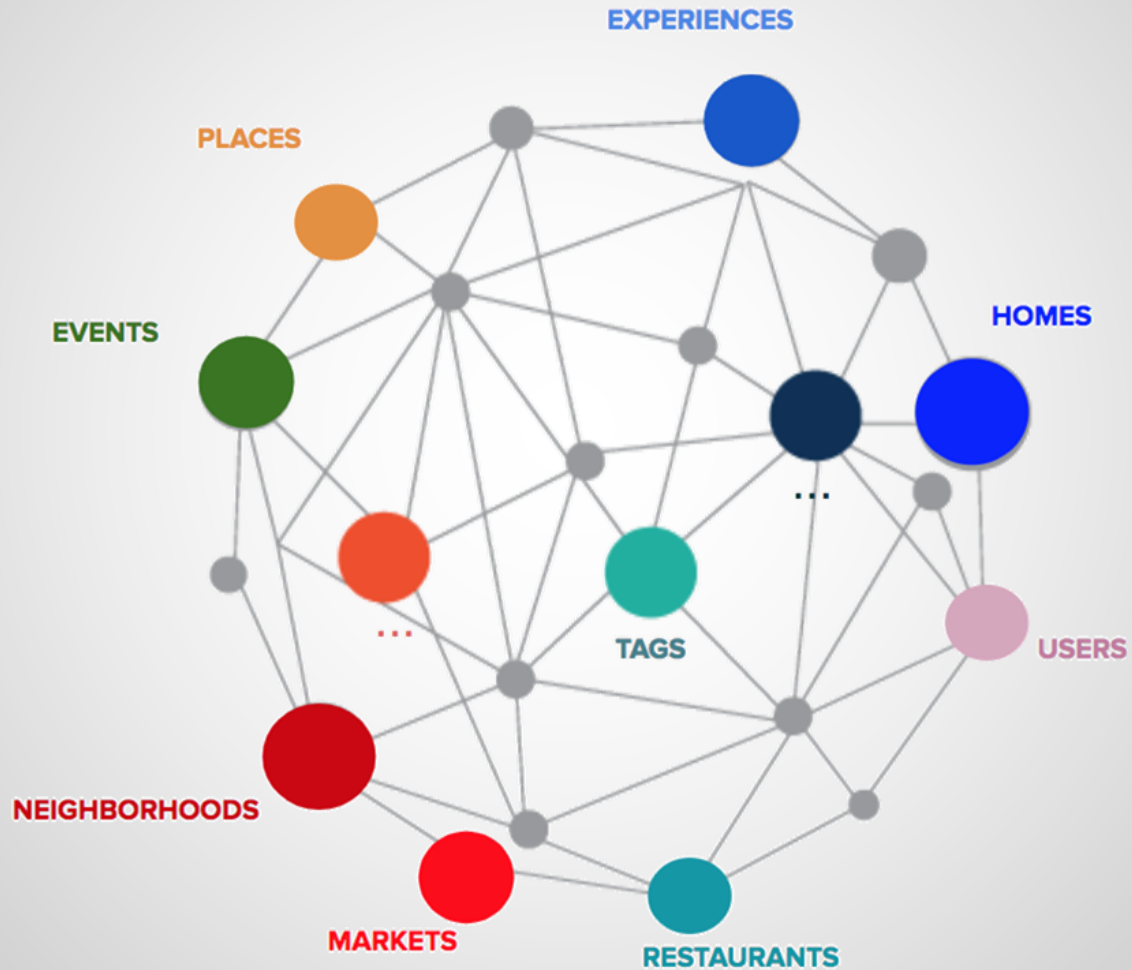
Thompson Reuters Information delivery

2 billion relationships linking shares, people, companies, events, and others

Oct 2017



Airbnb

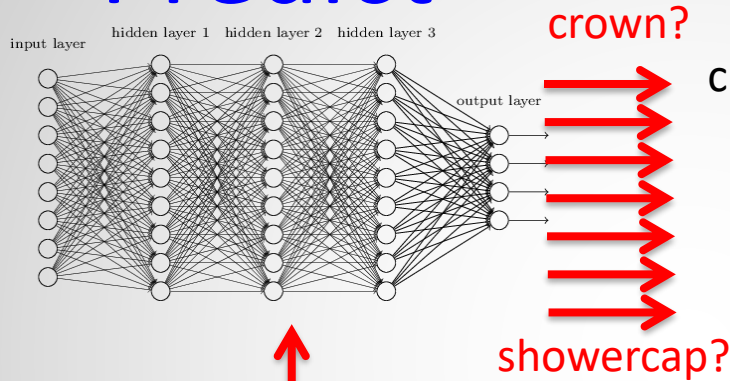


Research challenges in Knowledge Representation:

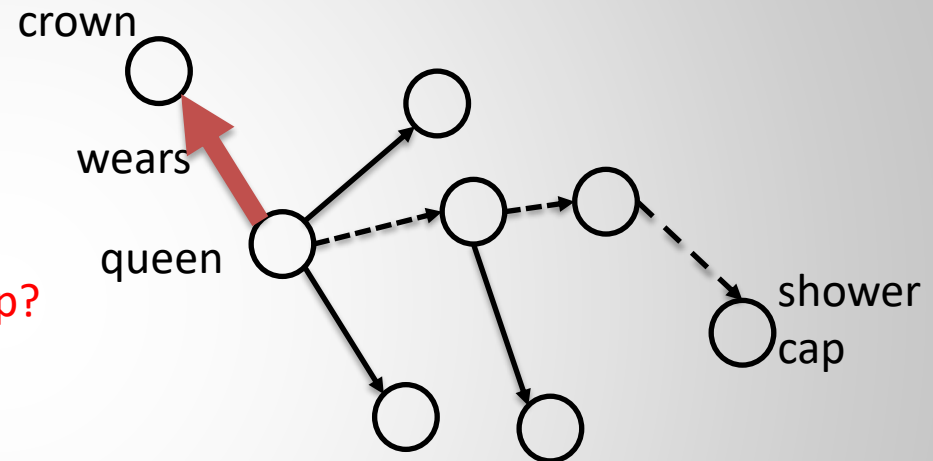
Neuro-symbolic systems,
combining the strengths of KR and ML,
“combining system 1 and system 2”
“AI systems that use all parts of our brain”

Informed ML using KR

Predict



Select

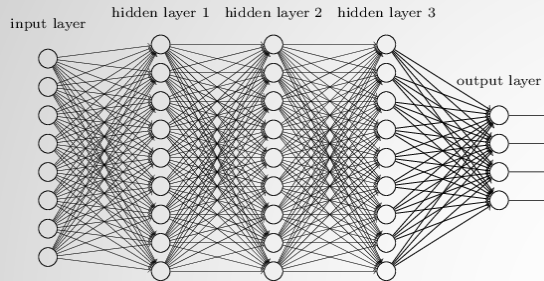


?

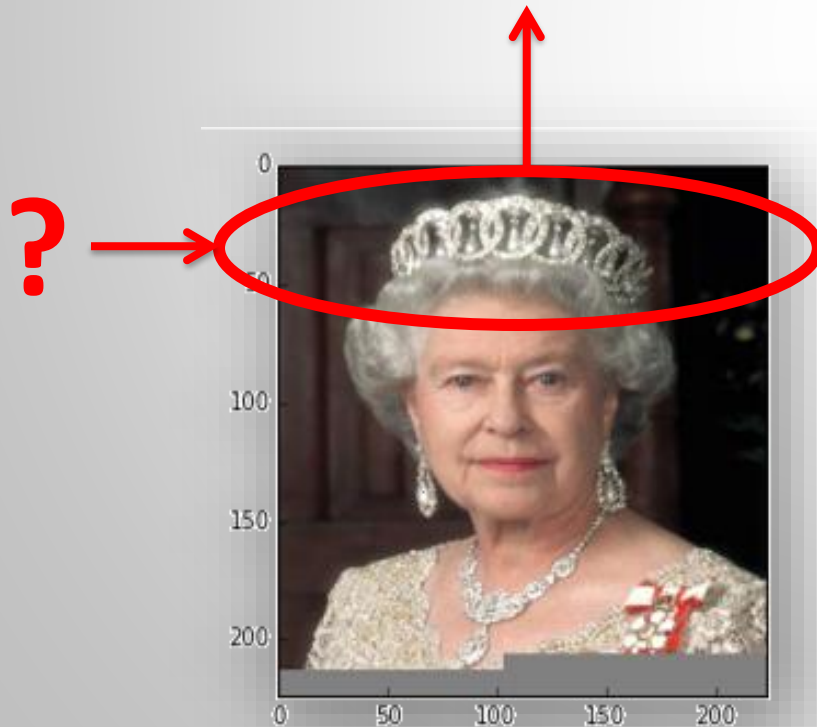


Explainable ML using KR

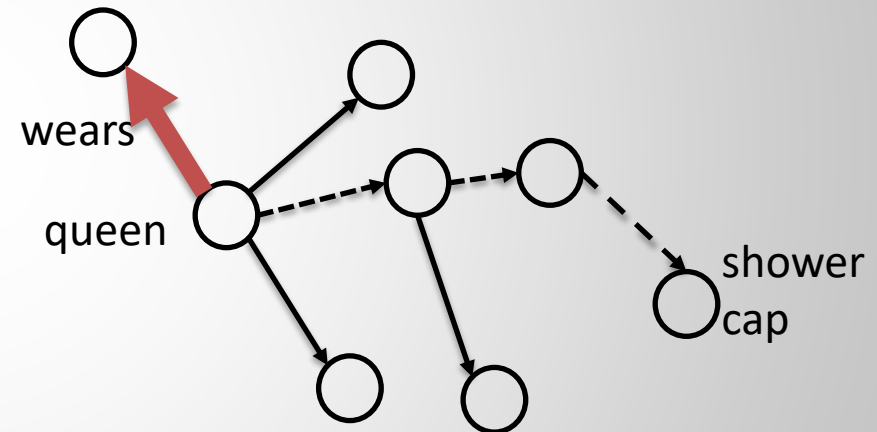
Predict



Explain



crown





Context aware ML using KR (informed ML)



flower?
cushion?

$P(\text{cushion} | \text{chair}) \gg P(\text{flower} | \text{chair})$

“Given the context of chair,
a **cushion** is much more likely
than a **flower**”

$\forall x, y \text{ chair}(x) \wedge \text{partOf}(y, x) \rightarrow$
 $\text{cushion}(y) \vee \text{armRest}(y)$

“Parts of a chair are: cushion and armrest”

Summary

- Two dominant lines of thinking in AI:
 - Statistical
 - Symbolic
- We looked at the symbolic school:
Knowledge Representation
- Modern knowledge bases can get very large
- They are in widespread practical use
- Big Question: how to combine the statistical and the symbolic school?
“AI that combines all parts of our brain”